# Sanaul Haque

## Rokib_Al_Dhin_Raadh_Proposal

Assignment 108

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# Can a Machine's Own Failure Record Be Read?

## Evaluating the Human Usability and Reliability of Automatically Generated Neural Network Incident Logs

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**Research Topic:** Evaluating whether automatically generated neural network failure records are usable by human engineers, reliable when the automated monitor errs, and sufficient for training-time record-keeping obligations under the European Union Artificial Intelligence Act.

## Research Question

Can automatically generated neural network failure records help engineers diagnose faults more accurately and faster than standard training logs?

This study addresses the following specific questions:

1. Does a structured telemetry report, an automatically generated log of each component's health, improve diagnostic accuracy and reduce time-to-correct-diagnosis among machine learning practitioners aged 18 and above, against a control group given only loss curves and ordinary training logs?
2. How often does the automated health monitor suppress network components that were in fact working correctly, and how does this error rate change as the model's overall output layer, which the monitor trusts as its reference, becomes less accurate?
3. Which fields in the telemetry record satisfy the training-time record-keeping obligations of Article 12 of the European Union Artificial Intelligence Act, and which obligations can a record of this kind not satisfy?
4. Is targeted diagnosis genuinely responsible for the network's recovery, or can the same recovery be produced by correcting randomly chosen components instead?

The independent variable is the form of failure record shown to a diagnostician; the dependent variables are diagnostic accuracy, time to correct diagnosis, and false-alarm rate. I hypothesise that the telemetry report will cut time to correct diagnosis and raise accuracy against the control, because it localises the fault to a named component; that false suppression will rise as the classifier's confidence becomes less well matched to its accuracy, because the gate is anchored to that classifier; and that random correction will recover significantly less than targeted correction. A null or reversed result refutes each.

## Introduction

Deploying an artificial intelligence system responsibly means being able to say, when it fails, which part of it failed and why. Standard neural network training makes that impossible in principle: because a single error signal is spread backwards across every parameter at once, a component that dies or destabilises leaves no timestamp, no identifier, and no record of what was done about it [1], [2]. The opacity is a by-product of how the network is trained, not something inherent to deep learning.

This is now a legal problem as well as a technical one. Article 12 of Regulation (EU) 2024/1689, the European Union's Artificial Intelligence Act, requires high-risk AI systems to automatically record events across their lifetime, and the NIST AI Risk Management Framework asks developers to document system behaviour throughout development [3], [4]. Yet no technical standard defines what such a record should contain.

In prior work I showed that the record can be produced [5], and a companion analysis of when this self-repairing training still converges has been accepted at DeepMath 2026 [6]. A training method called ORMAS gives every component its own diagnostic signal, making each component's health individually measurable and letting the network detect and repair its own failures while logging every intervention. Across 383 controlled experiments on one consumer graphics card, this produced roughly 340 million logged events. After a layer was deliberately destroyed partway through training, the network repaired itself to $80.3 \pm 1.6\%$ accuracy, while an equally sized

standard network stayed at $10.0 \pm 0.0\%$, random guessing.

However, the case for building such a record rests on an assumption that has never been tested: that a human engineer can actually read one and act on it. None of those 340 million events was ever shown to a person, no diagnosis time was measured, and no false-alarm rate was established. The record is also not always correct. On one test image, two components had correctly identified it, with 68.8% and 74.8% confidence, yet the monitor suppressed both because they disagreed with the network's overall output layer, which was itself wrong. It suppressed the correct signal, an error in the most dangerous direction.

The urgency is that the legal duty already exists while the evidence does not: systems are being certified against a record nobody has shown a human can use. What follows is therefore unusual in attacking a claim my own earlier work made rather than defending it. It treats the audit record itself as the object under investigation, asking whether that record is usable by humans, reliable when the monitor errs, and sufficient for the obligations it is said to satisfy.

## Literature Review

Most interpretability research treats a trained model as a finished object examined from outside. LIME [7] and SHAP [8] estimate why a model made a prediction after the fact, but Adebayo *et al.* showed such methods give similar outputs even when the model's parameters are randomised, failing basic reliability checks [9]. Rudin argues high-stakes work should instead use models interpretable by design [10]. This literature concerns what a decision means, not which component broke and when.

A second area studies training failures directly. Gradient conflict can be reduced by projection methods such as PCGrad [11], and local learning rules avoid global backpropagation altogether [12]. These prevent failure but leave no durable, component-level record of what occurred.

A third area evaluates explanations with real users. Poursabzi-Sangdeh *et al.* found more transparent models did not reliably help people detect mistakes, and sometimes made them worse [13]. Kaur *et al.* found data scientists over-trusted interpretability tools and misread their output [14]. Hase and Bansal measured whether explanations help users predict model behaviour [15]. All evaluate explanations of single predictions. None evaluates a training-time failure record against a deliberately injected fault where the answer is known in advance.

Finally, governance research specifies what documentation should exist, through model cards [16], datasheets [17], and the AI Act's record-keeping duties [3], but stops at the document level and does not say which automatically generated fields satisfy which obligation.

**Gap in the existing literature.** Prior work has produced component-level telemetry [5], tested explanations with human participants [13]–[15], and defined documentation obligations [3], [16], but no study has joined the three. None measures whether an automatically generated training-time failure record improves human diagnosis under known ground truth, establishes how often the monitor producing it is itself wrong, or maps its fields to named legal obligations. This study fills that gap.

## Research Methodology

This study uses a mixed-methods design: usability is a claim about people and cannot be settled from log files, while the reliability and causation questions need controlled interventions on the system itself.

**Fault Corpus:** I will generate 120 training runs on the CIFAR-10 and CIFAR-100 image datasets using my existing codebase, deliberately injecting faults across the seven failure types the system recognises: dead, exploded, saturated, oscillating, loss-stagnant, gradient-dead and low-confidence. I will record each run twice: once as a structured telemetry report, and once as a conventional record of loss curves, accuracy curves and ordinary logs. Because faults are injected on purpose, the correct diagnosis is known in advance, so answers can be scored right or wrong rather than self-reported.

**Quantitative Phase:** I will recruit sixty participants: machine learning practitioners aged 18 and above with at least one year of practical experience, reached through open-source communities, university machine learning society mailing lists and practitioner forums. Because the population is defined by expertise rather than geography, recruitment is international and online. I will screen participants with a short competency task and assign them at random to the telemetry or control group. Each will diagnose eight runs in counterbalanced order, giving about 480 trials. For each run, participants identify the failing component, name the failure type, and rate their confidence. The measures are diagnostic accuracy against the injected fault, time to correct diagnosis, false-alarm rate on fault-free runs, and perceived workload using the NASA Task Load Index [18].

**Qualitative Phase:** I will interview twelve participants, selected by purposive sampling, about how they read the report, which parts they relied on, and where it misled them, then transcribe and analyse the interviews by thematic analysis following Braun and Clarke [19].

**Computational Phase:** Two further experiments run without human participants. First, I will vary the accuracy of the network's global classifier systematically across 60 runs by controlling label noise and training length, and measure the rate at which the monitor suppresses correctly functioning components against that accuracy. Second, a three-way comparison will test whether targeted diagnosis matters: full diagnosis-driven correction, correction disabled, and correction applied to random components at identical rate and strength. Separately, each telemetry field will be mapped against Article 12 of Regulation (EU) 2024/1689 and the NIST framework by two independent coders, myself and one trained volunteer, with agreement reported using Cohen's κ.

**Data Analysis:** I will analyse diagnostic accuracy using mixed-effects logistic regression, with participant and run as random effects, since each participant contributes several trials, and time to diagnosis using survival analysis so that trials which time out are handled correctly. False-alarm rates will be compared using a two-proportion test, and the three-way computational comparison using one-way ANOVA across three seeds per condition. A simulation-based power analysis assuming a moderate effect size and eight trials per participant indicates that 48 participants give approximately 80% power; the target of 60 allows for dropout. Interview findings will be integrated with the numerical results by triangulation, so the qualitative accounts help explain why the measured differences occurred.

## Expected Outcomes

I expect four results: a measured effect of the report on human diagnosis, as differences in accuracy, median time to correct diagnosis and false-alarm rate with confidence intervals; a curve showing how often the monitor silences correctly working components as its reference classifier degrades, establishing when the record should not be trusted; a conformance matrix mapping each telemetry field to the Article 12 obligations it satisfies and those it cannot; and a verdict on whether targeted diagnosis is necessary for recovery.

## Project Practicalities

I have planned the project over ten months: two to build the fault corpus and pilot the task, three for recruitment and data collection, three for the computational experiments, and two for analysis and writing. All computational work runs on one consumer NVIDIA RTX 3090, the hardware that produced the 383 earlier experiments, so the study needs no institutional facilities.

Participation is voluntary and minimal-risk: adults perform a short diagnostic task on machine-generated logs, and no personal data is collected. I will obtain informed consent, participants may withdraw at any time, no identifying data will be kept beyond a pseudonymous identifier, and the injected faults will be explained at debriefing. Ethical review will be sought where an affiliating body is available; otherwise the protocol follows established guidance for minimal-risk internet research.

## Roadblocks and Potential Limitations

Several genuine difficulties should be stated plainly. The most serious is recruitment. Sixty practitioners willing to spend forty minutes on an unpaid task is an ambitious target for an independent secondary-school researcher without a laboratory or funding, and the realistic risk is falling short. If fewer than forty are obtained, I will report the human experiment as a pilot study with wider confidence intervals rather than a confirmatory result, and the computational experiments, which need no recruitment, will carry the main findings.

Second, participants recruited from open-source communities will not represent all practitioners; they are likely to be more experienced and more interested in interpretability than average, which may make the report look more useful than it would be in practice. I will record experience levels and report this.

Third, injected faults are cleaner than those arising naturally in deployment, so good performance here does not guarantee it on real incidents. Fourth, the study covers small image-classification networks only and may not transfer to large language models. Fifth, the legal mapping is one researcher’s interpretation, not legal advice. Finally, I am the author of the system being evaluated, which is a clear risk of bias; to reduce it, the diagnostic task will be scored automatically against the injected faults, and the analysis plan published before any data is collected.

## Conclusion

Transparent architectures are justified on the grounds that they produce a record a human can read. That claim has never been tested on a human. This project tests it directly, under conditions where the correct answer is known in advance, and on a system I built myself, which is precisely why a negative result would be as publishable as a positive one. It also establishes when the monitor is itself untrustworthy, and what its record can and cannot prove to a regulator. Either way, a field currently building on an assumption gets evidence instead.

## Post-Program Plan

After completion, I will submit the results to the Youth Research Journal and to a peer-reviewed venue concerned with trustworthy machine learning. The fault corpus, telemetry reports, analysis code and anonymised responses will be released publicly so others can reuse the materials and repeat the study. I will also write a plain-language summary for the practitioners who build and monitor these systems. Beyond this project, I intend to carry the work to university and to larger models, where undetected failure matters far more.

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